

SIEM - ANALYSIS USING SPLUNK

Case 1 : Bruteforce Analysis

You've just started your first shift as a SOC analyst at an MSSP. Only a few minutes have passed since an alert about a possible brute force attack appeared on the platform.

Alert Details:

Alert Name: Brute Force Activity Detection

Time: 17/09/2025 9:00:21 AM

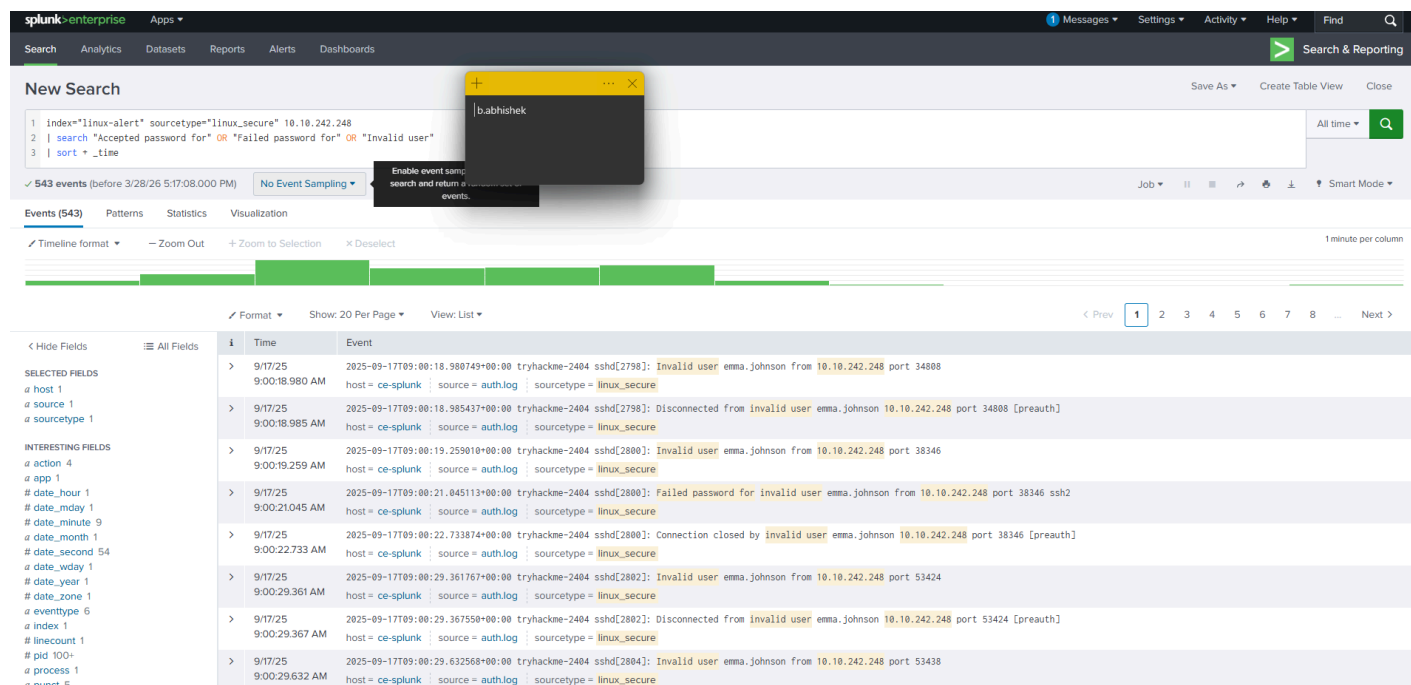
Target Host: WIN-2404

Source IP: 10.10.242.248

Your job is to investigate this activity and decide whether it should be considered suspicious.

Query Used:

```
index="linux-alert" sourcetype="linux_secure" 10.10.242.248
| search "Accepted password for" OR "Failed password for" OR "Invalid user"
| sort + _time
```

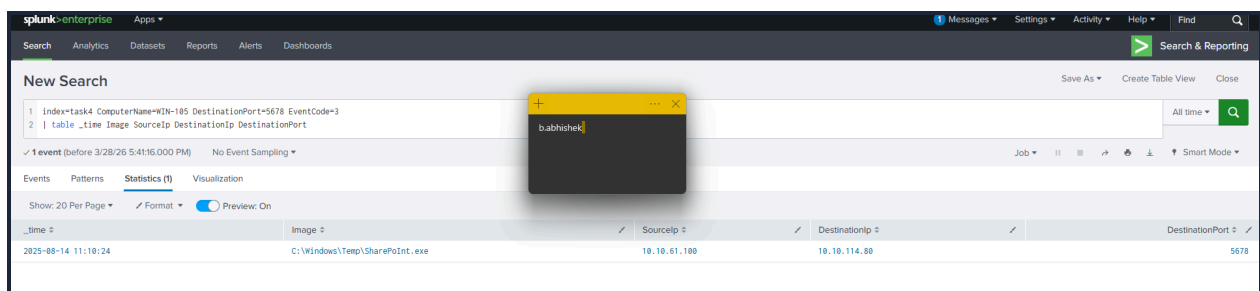


Case 2 - Log Based Scenario using Sysmon and WinEvent Logs

You are an SOC Level 1 Analyst on shift and have received an alert indicating a suspicious network connection using port 5678 on the WIN-105 host. Your task is to conduct an investigation and determine whether this activity is suspicious.

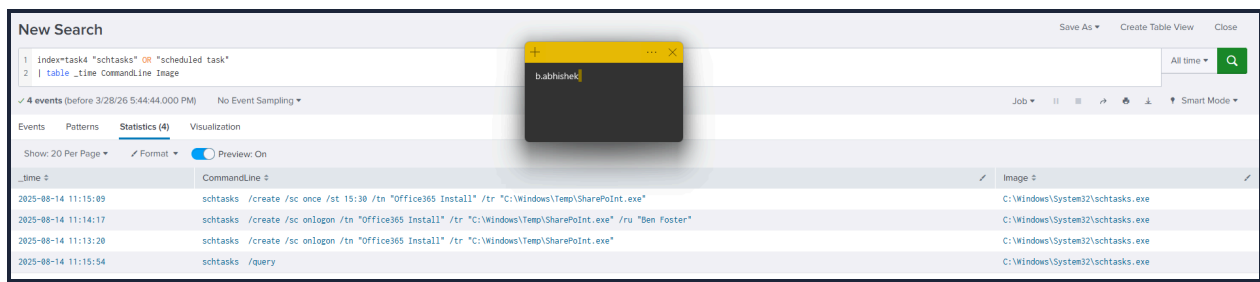
Query Used:

**index=task4 ComputerName=WIN-105 DestinationPort=5678 EventCode=3
| table _time Image SourceIp DestinationIp DestinationPort**



Query Used:

index=task4 "schtasks" OR "scheduled task"
| table _time CommandLine Image



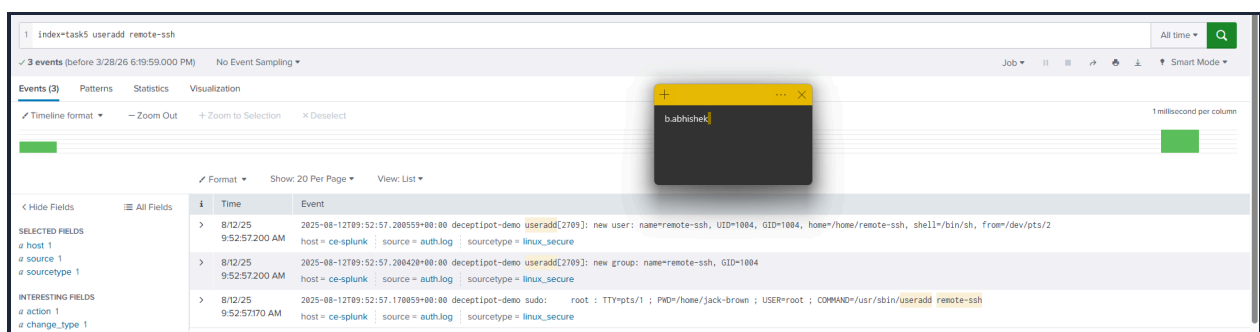
Case 3 :Remote ssh persistence

You are an **SOC Level 1 Analyst** on shift and have received an alert indicating possible **persistence** through the creation of a new **remote-ssh** user on an Ubuntu server.

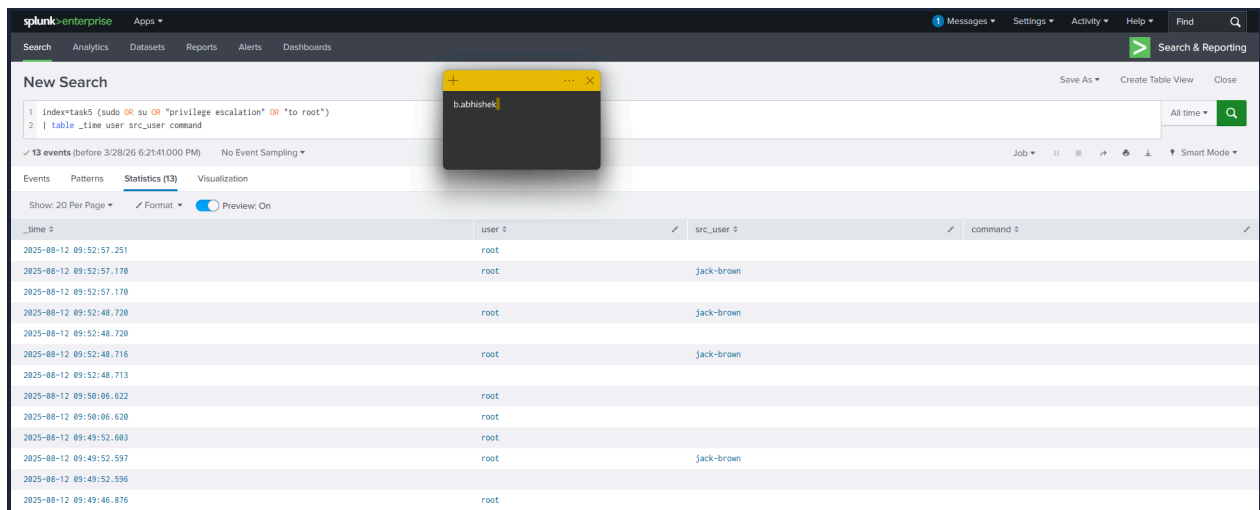
Your task is to dive into the logs and determine exactly what happened on the system.

Query Used:

index=task5 useradd remote-ssh



**index=task5 (sudo OR su OR "privilege escalation" OR "to root")
| table _time user src_user command**



New Search

1 index=task5 (sudo OR su OR "privilege escalation" OR "to root")
2 | table _time user src_user command

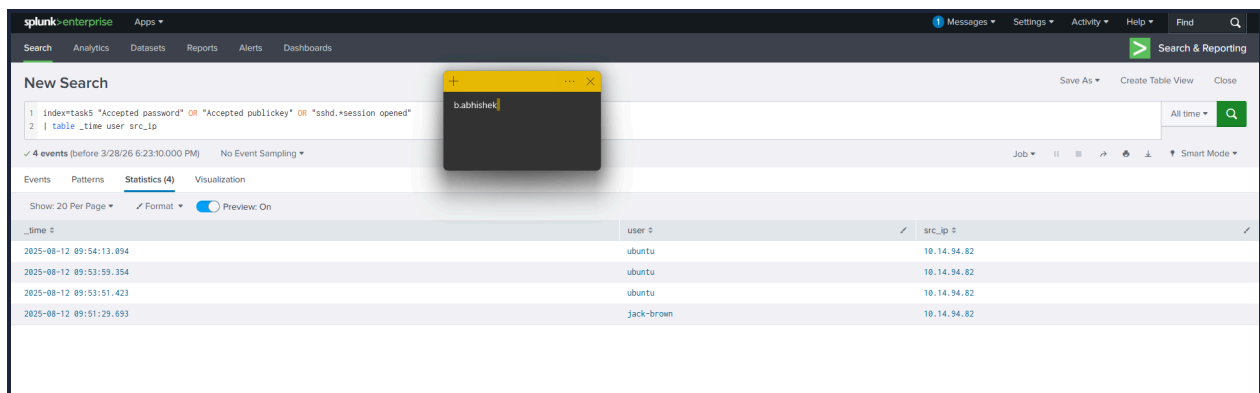
✓ 13 events (before 3/28/26 6:21:41.000 PM) No Event Sampling

Events Patterns **Statistics (13)** Visualization

Show: 20 Per Page ✓ Format Preview: On

_time	user	src_user	command
2025-08-12 09:52:57.251	root		
2025-08-12 09:52:57.170	root	jack-brown	
2025-08-12 09:52:57.170			
2025-08-12 09:52:48.720	root	jack-brown	
2025-08-12 09:52:48.720			
2025-08-12 09:52:48.716	root	jack-brown	
2025-08-12 09:52:48.713			
2025-08-12 09:50:06.622	root		
2025-08-12 09:50:06.620	root		
2025-08-12 09:49:52.683	root		
2025-08-12 09:49:52.597	root	jack-brown	
2025-08-12 09:49:52.596			
2025-08-12 09:49:48.876	root		

**index=task5 "Accepted password" OR "Accepted publickey" OR "sshd.*session
opened"
| table _time user src_ip**



New Search

1 index=task5 "Accepted password" OR "Accepted publickey" OR "sshd.*session opened"
2 | table _time user src_ip

✓ 4 events (before 3/28/26 6:23:10.000 PM) No Event Sampling

Events Patterns **Statistics (4)** Visualization

Show: 20 Per Page ✓ Format Preview: On

_time	user	src_ip
2025-08-12 09:54:13.894	ubuntu	10.14.94.82
2025-08-12 09:53:59.354	ubuntu	10.14.94.82
2025-08-12 09:53:51.423	ubuntu	10.14.94.82
2025-08-12 09:51:29.693	jack-brown	10.14.94.82

**index=task5 "Failed password" OR "authentication failure"
| stats count**

The screenshot shows the Splunk Search interface with a search query: `index=task5 "Failed password" OR "authentication failure" | stats count`. The search results show 7 events. A terminal window is overlaid on the search bar, showing the command `babhishek`.

count
7

**index=task5 (cron OR crontab OR "reverse shell" OR nc OR netcat OR "ssh -R")
| table _time command**

The screenshot shows the Splunk Search interface with a search query: `index=task5 (cron OR crontab OR "reverse shell" OR nc OR netcat OR "ssh -R") | table _time command`. The search results show 8 events. A terminal window is overlaid on the search bar, showing the command `babhishek`.

_time	command
2025-08-12 10:00:32.041	
2025-08-12 10:00:30.396	
2025-08-12 10:00:01.270	
2025-08-12 09:59:34.245	
2025-08-12 09:57:35.272	
2025-08-12 09:57:35.268	
2025-08-12 09:56:33.572	
2025-08-12 09:55:01.259	

